BK boundary: actual torsion field versus roots of unity

Actual result. For E/F7: $y^2=x^3+3x+2$, Lean proves $\Delta\neq 0$, $\#E(F7)=9$ and $\#E[3](F7bar)=9$. For every positive k and every compatible embedding of a field of size 7^k, all geometric 3-torsion is rational exactly when 3 divides k. Yet 3 divides 7^k−1 for every k; the cyclotomic embedding degree is 1. Signed standing: unverified. [pcc:unverified]

Positive extension degree k	1	2	3	4	5	6	7	8	9	10	11	12
3 divides 7^k−1	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
All geometric 3-torsion rational	no	no	yes	no	no	yes	no	no	yes	no	no	yes

Figure. These are corollaries of the all-k Lean theorem, not sampled measurements. The entire geometric torsion first becomes rational in degree 3; the rows disagree for infinitely many degrees, including every 3n+1. [pcc:unverified]

Coordinate proof. The actual point/S3 equivalence factors the literal division quartic as $3(x-5)(x^3+5x^2+3x+2)$. The rational branch has $y=\pm 3$. On the cubic branch, $y=\pm 3x$, $x^7=6x^2+6x+3$, and $(6x^2+6x+3)^2=x+1$. Explicit small polynomial factors and (7:F7[X])=0 prove the identities by checked ring arithmetic. They give actual point Frobenius cubed equal to the identity on all 3-torsion. [pcc:unverified]

All extensions. The existing actual nonfixed geometric point forces period 3. Modular decomposition of every k then gives fixedness iff 3 divides k. The existing proved finite-field descent connects those actual Frobenius iterates to actual point-map images, for every chosen field and compatible embedding. GaloisField7k cardinality and algebraic-closure embeddings are constructed. The nine-point geometric count follows from the existing arbitrary short-curve theorem after checking the actual curve and discriminant. No torsion basis, determinant, Weil pairing or rationality conclusion is assumed. [pcc:unverified]

Source correction. Original BK1998 Theorem1, pp142–143, explicitly excludes r dividing q−1. Its Remarks already warn that r^2 dividing #E does not repair deleting this hypothesis, and give a different F19 example. Our actual F7 certificate confirms that excluded-boundary mechanism. It disproves the unrestricted corpus reading; no qualified source theorem or novelty claim is asserted. [pcc:unverified]

Checks. Whole-project gate 32/0failures/0trusted. Lean4.35.0 / Mathlib56a00585…. Only standard foundations. No holes, custom axioms, native evaluator, raised limits, global elaboration flags or waivers. Failed source/log/exit records are retained. Exact complete main type and all dependency/artifact hashes are in sources.json; see proof-map.json and evidence/. [pcc:unverified]

Scope. No native experiment, paid compute or agent upload. KB unavailable; original primary paper, existing actual F7/Frobenius/cardinality modules and pinned Mathlib reviewed. Generic all-prime rank/determinant/Weil bridges and the complete all-extension census remain open in the fourteen-item map. This directory does not assert a separately computed Frobenius matrix. [pcc:unverified]